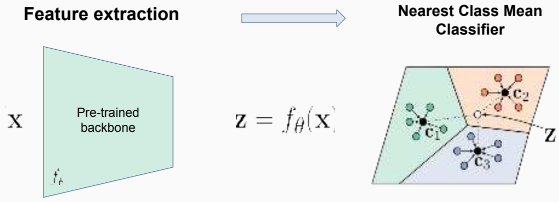
A Statistical Model for Predicting Generalization in Few-Shot Classification

IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Yassir Bendou[1], Bastien Pasdeloup[1], Giulia Lioi[1], Vincent Gripon[1], Lukas Mauch[2], Stefan Uhlich[2], Fabien Cardinaux[2], Ghouthi Boukli Hacene[2,3] and Javier Alonso Garcia[2]
[1]IMT Atlantique, Lab-STICC, Brest, France
[2]Sony Europe, R&D Center, Stuttgart Laboratory 1, Germany
[3]MILA, Montréal, Canada

Introduction 1

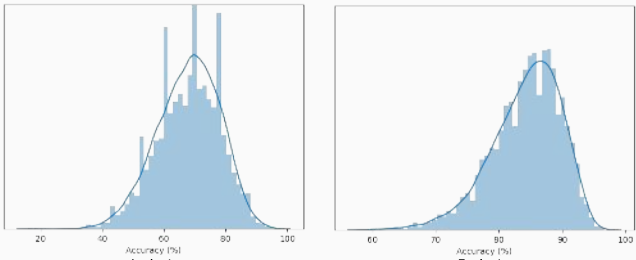
- Deep learning has known a large success due to its capacity to exploit large training datasets.
- Few-shot learning is a setting where training data is limited (satellite imaging, fMRI, seismic activity...).



Classifical Few-shot pipeline

Scientific Challenge 2

- Even when sampled from the same dataset, the performance of the classifier strongly depends on the choice of the shots:

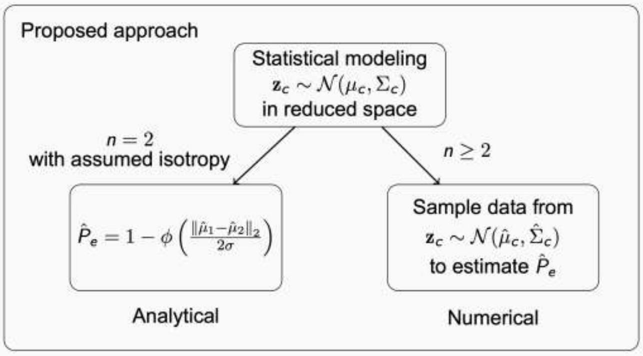


Histogram of accuracies on MiniImageNet using 10k runs

- How to approximate the generalization in a few-shot setting where a validation set is not available?

Our approach 3

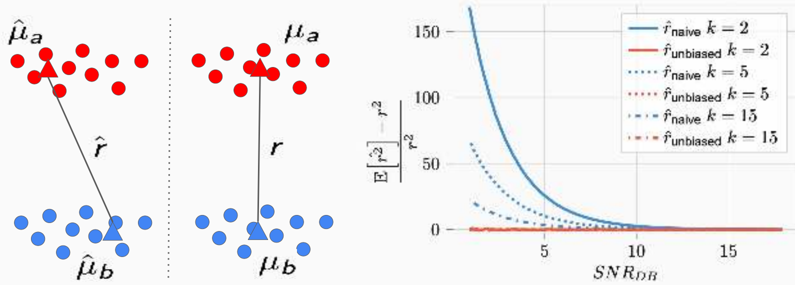
- Due to limited amount of available samples, we use a simple mathematical model with few parameters in order to infer the accuracy.
- We assume that each class in the feature space is a multivariate Gaussian distribution.



Key findings 4

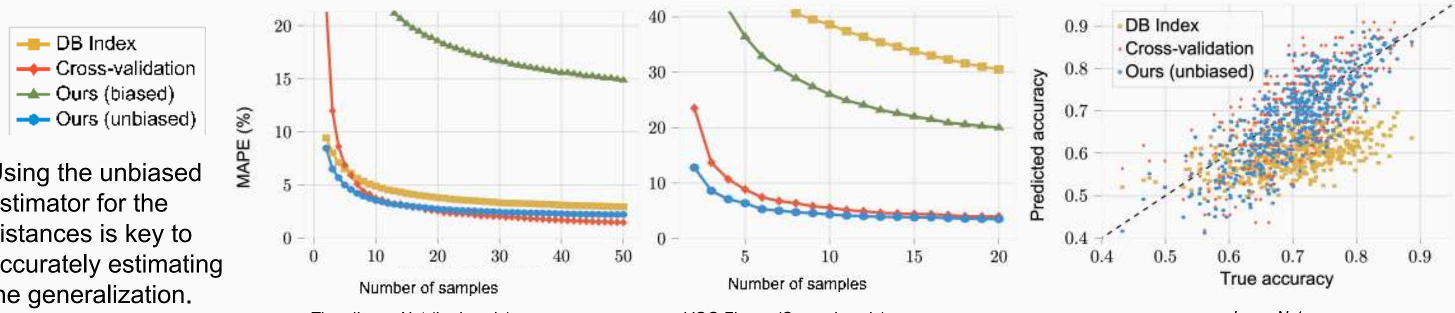
Let $r = \|\mu_a - \mu_b\|_2$ be the distance between the class centers.
The naive estimator of the distances between class centers is biased:

$$\mathbb{E}_{\substack{a \sim p_a \\ b \sim p_b}}\left(\hat{r}^2\right) - r^2 = \frac{\mathrm{Tr}\left(\Sigma_a + \Sigma_b\right)}{k}$$



Bias of the distance estimator for different SNRs and number of shots.

Results 5



Using the unbiased estimator for the distances is key to accurately estimating the generalization.

Mean-Absolute-Percentage Error of different generalization predictors against the number of samples on 1000 5-class classification problems.

Scatter plot of 5-class few-shot problems with 10 samples per class from ImageNet

Conclusion 6

Takeaways: 1) Predicting generalization in few-shot settings is an interesting topic for many applications.
2) We propose a first baseline with plenty of open questions (Gaussian assumption, fewer hyperparameters).

To contribute: Link with materials to run the experiments: https://github.com/ybendou/fs-generalization

Full paper link